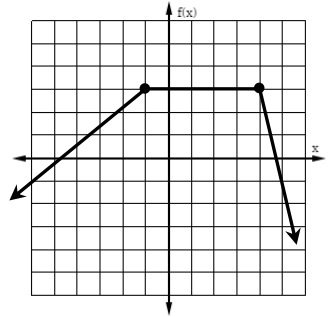
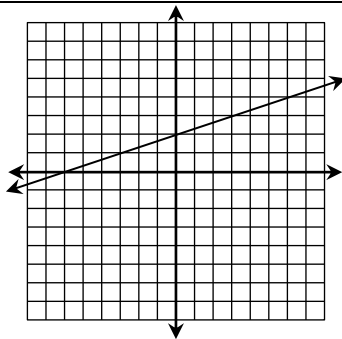
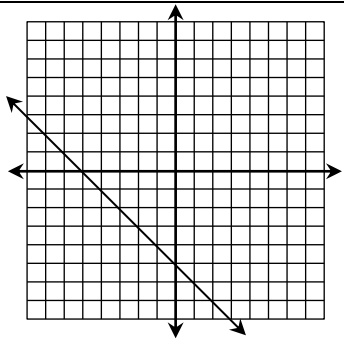
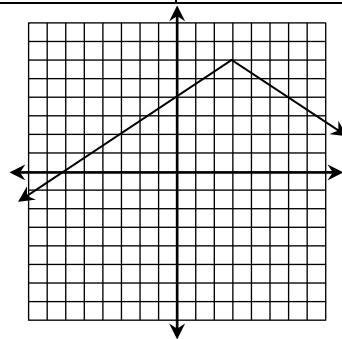
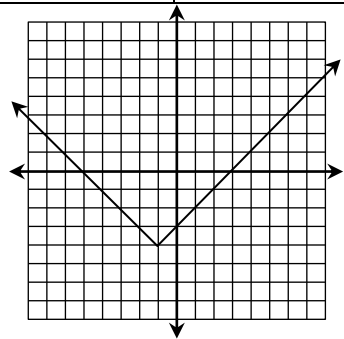
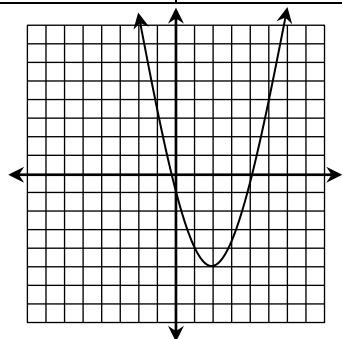
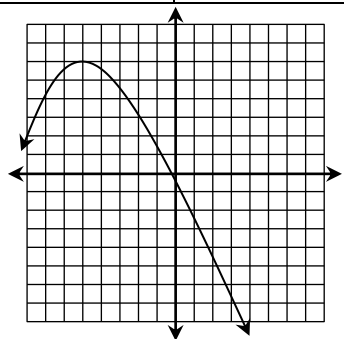


Name:	Date:
Topic:	Class:

Main Ideas/Questions	Notes/Examples			
INCREASING & DECREASING Intervals	Moving from left to right, a function may increase, decrease, or remain constant on certain intervals. For example, the function to the right is increasing on the interval _____. constant on the interval _____. decreasing on the interval _____.			
	EXAMPLES	Identify the intervals on which each function increases or decreases.		
1. 		2. 		
Increasing Interval(s): Decreasing Interval(s):		Increasing Interval(s): Decreasing Interval(s):		
3. 		4. 		
Increasing Interval(s): Decreasing Interval(s):		Increasing Interval(s): Decreasing Interval(s):		
5. 		6. 		
Increasing Interval(s): Decreasing Interval(s):		Increasing Interval(s): Decreasing Interval(s):		

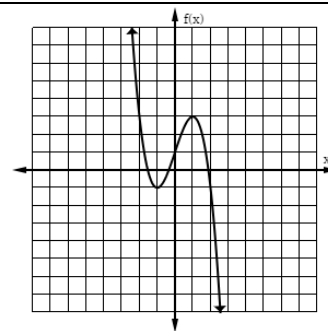
END BEHAVIOR

As x moves towards positive infinity and negative infinity, end behavior describes what is happening to the value of $f(x)$.

Describe the end behavior of the graph to the right:

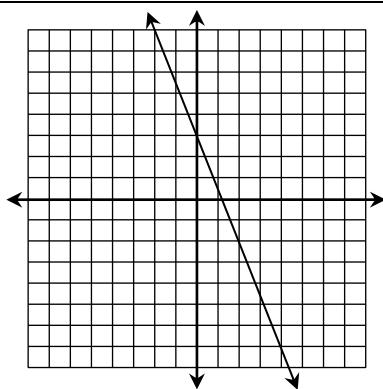
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____



Describe the end behavior of each function.

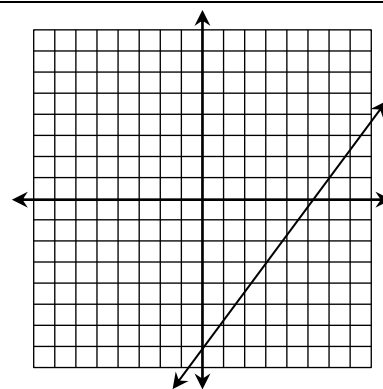
7.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

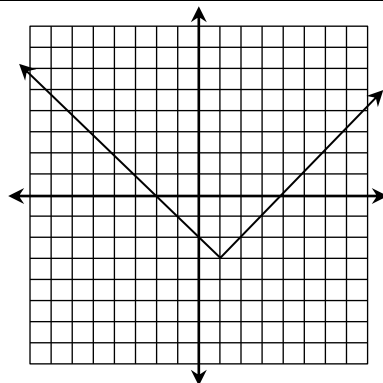
8.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

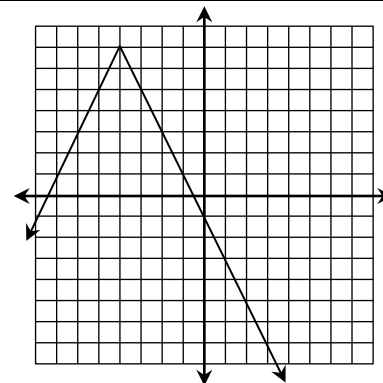
9.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

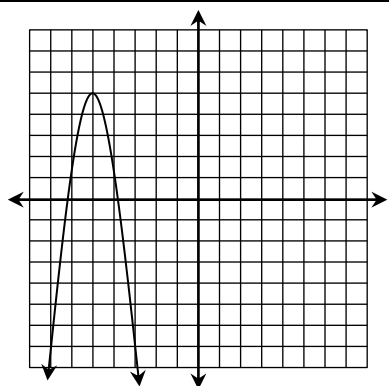
10.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

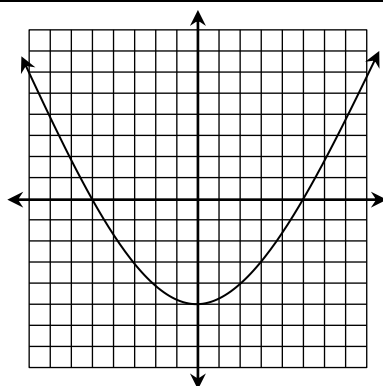
11.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

12.



As $x \rightarrow \infty$, $f(x) \rightarrow$ _____

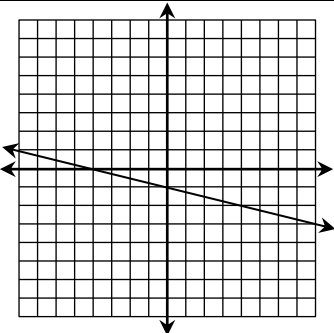
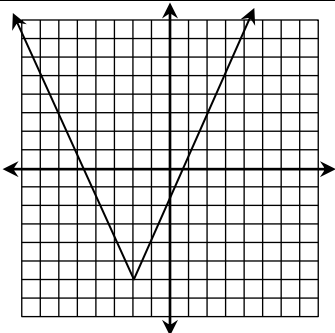
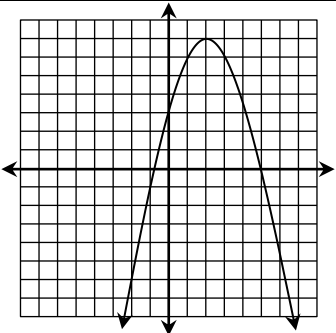
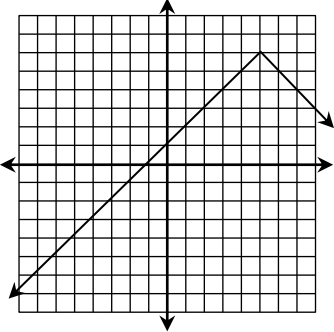
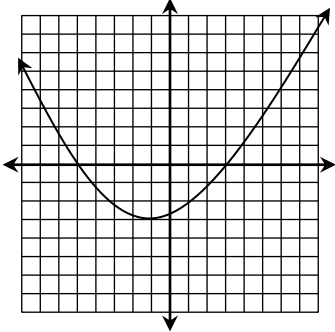
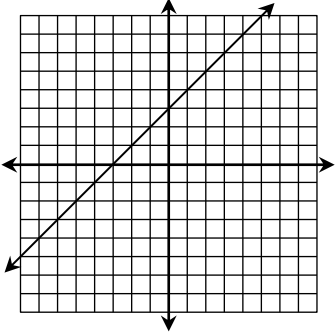
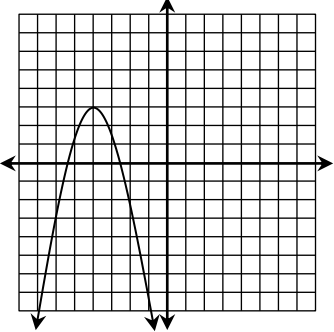
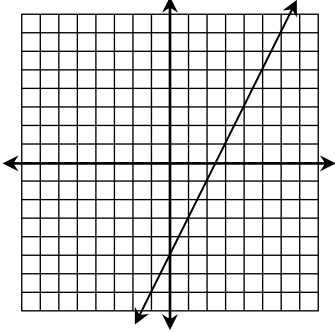
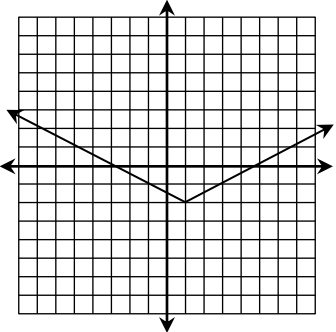
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____

Name: _____

Unit 3: Parent Functions & Transformations

Date: _____ Per: _____

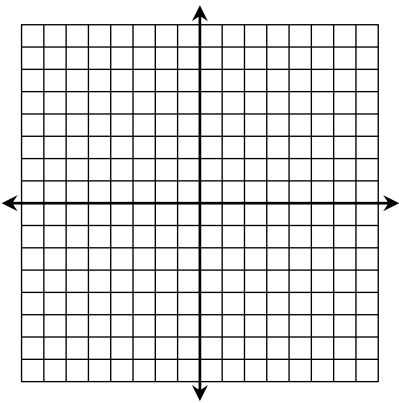
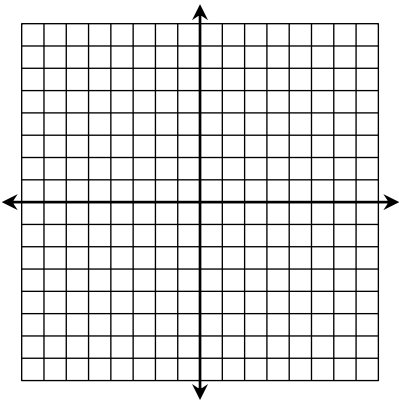
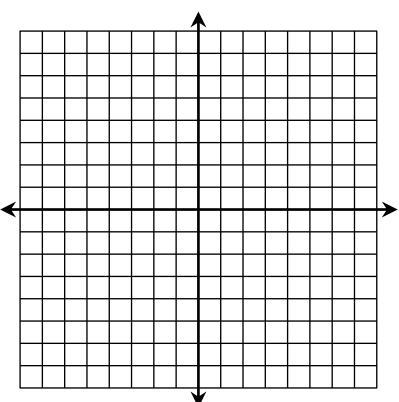
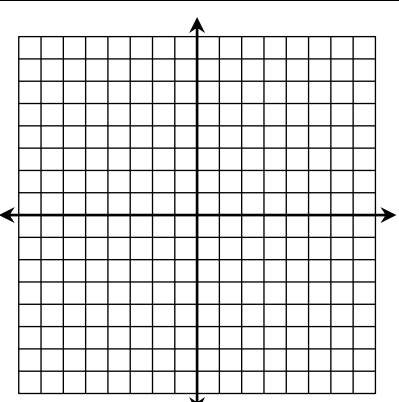
**Homework 8: Increasing/Decreasing Intervals
& End Behavior**

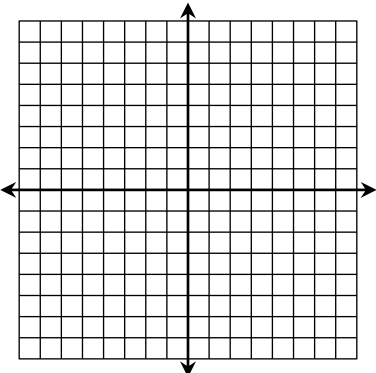
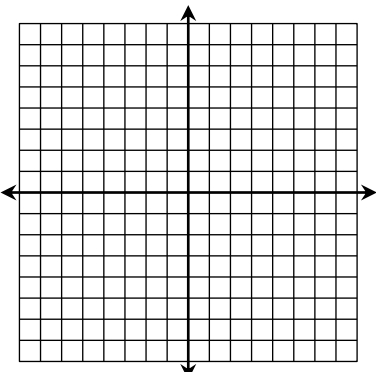
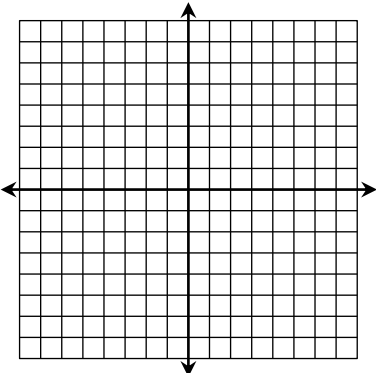
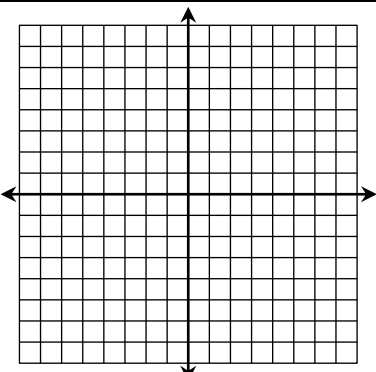
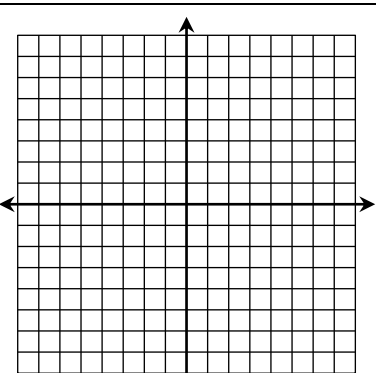
Identify the increasing and decreasing intervals and the end behavior of each graph.					
1. 		2. 		3. 	
Increasing Int(s):		Decreasing Int(s):		Increasing Int(s):	
				Decreasing Int(s):	
End Behavior:		End Behavior:		End Behavior:	
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____	
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____	
4. 		5. 		6. 	
Increasing Int(s):		Decreasing Int(s):		Increasing Int(s):	
				Decreasing Int(s):	
End Behavior:		End Behavior:		End Behavior:	
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____	
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____	
7. 		8. 		9. 	
Increasing Int(s):		Decreasing Int(s):		Increasing Int(s):	
				Decreasing Int(s):	
End Behavior:		End Behavior:		End Behavior:	
As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____		As $x \rightarrow \infty$, $f(x) \rightarrow$ _____	
As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____		As $x \rightarrow -\infty$, $f(x) \rightarrow$ _____	

PARENT FUNCTIONS

A **parent function** is the simplest function of the family.

A **family of graphs** displays similar characteristics.

PARENT FUNCTION	GRAPH	CHARACTERISTICS
LINEAR 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____
ABSOLUTE VALUE 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____
QUADRATIC 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____
CUBIC 		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Interval(s): _____

SQUARE ROOT		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____
CUBE ROOT		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____
RECIPROCAL		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____ • Asymptote(s): _____
EXPONENTIAL		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____ • Asymptote(s): _____
LOGARITHMIC		• D: _____; R: _____ • x-intercept: _____; y-intercept: _____ • End Behavior: As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ As $x \rightarrow$ _____, $f(x) \rightarrow$ _____ • Increasing Interval(s): _____ • Decreasing Intervals(s): _____ • Asymptote(s): _____